Customer Segmentation with K-Means++

ML | Implementation and interview guide

Purpose and intended demonstration

Discover customer groups and evaluate clustering stability using independent fits.

How the implementation works

Standardize features using training statistics. K-means++ chooses dispersed initial centers; Lloyd iterations alternate nearest-center assignment and mean updates. Empty centers retain their previous position.

Evaluation and evidence

Compare k=2,3,4,5 with a silhouette estimate on held-out rows. Report adjusted Rand index using synthetic labels solely as an external diagnostic, not for fitting.

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Setup and commands

Requires Python 3.10 or newer and NumPy. Create a virtual environment: python -m venv .venv. On Windows use .venv\Scripts\python.exe if activation is unavailable. Install with python -m pip install -r requirements.txt. Run python app.py, then open results/report.html in your browser. Run python -m unittest discover -v to execute tests. The default seed is 42; change it with --seed. Replace sample_input.json using --input path-to-file.json, preserving its schema. Algorithm-only projects generate synthetic arrays from the seed rather than consuming external files.

Architecture and file responsibilities

CLI arguments -> JSON input or seeded generator -> domain pipeline -> validation or evaluation -> JSON and HTML artifacts. Imported functions can be used in a notebook or a separate application. Model checkpoints preserve learned arrays where applicable; see app.py for parameter order. This package does not include a production web service. Each project is independent and includes its own helper code so the ZIP can be extracted and run alone. app.py holds the experiment and domain functions: fit, assign, silhouette, ari, run. kit.py contains reporting, input loading and only the retrieval or generation helpers needed by this project. sample_input.json contains editable inputs. test_app.py checks domain behaviors; test_reporting.py checks metrics and HTML output boundaries. results/metrics.json records outputs and results/report.html provides a browser-readable report. Numeric projects also save arrays or learned parameters to results/artifacts.npz; inspect its keys with numpy.load. MODEL_CARD.md explains intended use and limits. .github/workflows/tests.yml runs tests and the fixture experiment. No API key is needed.

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Limitations and next improvements

Silhouette favors compact Euclidean clusters. Segment IDs have no inherent business meaning; synthetic separation makes the example easy. Add mixed feature handling, bootstrap stability, segment profiles and a practical strategy for naming segments without using hidden labels.

Interview walkthrough

Present the problem and demonstrate one normal example, then one failure or boundary case. Explain the baseline or fixture mode, the split or validation design, and one limitation before discussing an extension. Be ready to trace data through the functions listed above. Explain why the chosen metric matches the task and why the available results cannot establish production performance. Add your own documented experiment and be precise about what you implemented or changed. These are AI-assisted educational projects; filenames and synthetic scores are not evidence of client work.

Verified build results

project

"Customer Segmentation with K-Means++"

selected k

3

validation silhouette

0.8090149858225745

external ari

1.0

profiles

[[-4.0530425384380955, -0.0044831536443812325], [3.995796443521654, -0.9399080048534569], [-0.022507751095259362, 3.9348943501386042]]

caveat

"The 200 held-out rows select k, so silhouette is validation performance, not an unbiased final test."

See results/metrics.json for all observations and TEST_RESULTS.txt for the executed tests. All datasets and replay responses shipped here are synthetic. Results can vary across seeds and numerical library versions.